

Circuit Crafter

Complete Component & Dev Board Library Reference

Every electrical, electronics & embedded component — category-wise

Legend

ALREADY BUILT (green bold) = component already in Circuit Crafter HIGH PRIORITY (amber bold) = most commonly used, implement first Normal = standard addition for future versions

Part 1: Electrical & Electronics Components — 28 categories, 557 components Part 2: Development Boards & Modules — 27 boards across 8 categories, 404+ pins documented

PART 1 Electrical & Electronics Components

Passive Components (28 components)

[EXISTS] Resistor	[EXISTS] Capacitor	[EXISTS] Inductor
[PRIORITY] Variable resistor (rheostat)	[PRIORITY] Potentiometer	[PRIORITY] Trimmer / trim pot
[PRIORITY] LDR (light-dependent resistor)	[PRIORITY] Thermistor (NTC/PTC)	[PRIORITY] Varistor (MOV)
Memristor	[PRIORITY] Electrolytic capacitor	[PRIORITY] Tantalum capacitor
[PRIORITY] Variable capacitor	Supercapacitor / ultracapacitor	[PRIORITY] Crystal / quartz resonator
Ceramic resonator	MEMS resonator	[PRIORITY] Fuse
[PRIORITY] PTC resettable fuse	[PRIORITY] Ferrite bead	[PRIORITY] Common-mode choke
[PRIORITY] Transformer (ideal)	Coupled inductors	Balun
Tapped inductor	RF coil	Air-core inductor
Saturable core reactor		

Diodes & Rectifiers (22 components)

[PRIORITY] Signal diode (1N4148)	[PRIORITY] Rectifier diode	[PRIORITY] Zener diode
[PRIORITY] Schottky diode	[PRIORITY] TVS diode (transient voltage suppressor)	[PRIORITY] Varactor / varicap diode
PIN diode	Tunnel diode	Gunn diode
IMPATT diode	Step-recovery diode	[PRIORITY] LED (light-emitting diode)
[PRIORITY] Infrared LED	Laser diode	[PRIORITY] Photodiode
Avalanche photodiode	[PRIORITY] Solar cell / photovoltaic cell	Shockley diode
Diac	[PRIORITY] Full-wave bridge rectifier	Ideal diode (SPICE)
Constant-current diode		

Bipolar Transistors (BJT) (8 components)

[PRIORITY] NPN BJT	[PRIORITY] PNP BJT	[PRIORITY] Darlington pair (NPN)
Darlington pair (PNP)	Sziklai pair (complementary Darlington)	[PRIORITY] Photo-transistor
[PRIORITY] NPN power transistor	PNP power transistor	

Field-Effect Transistors (FET) (16 components)

[PRIORITY] N-channel JFET	P-channel JFET	[PRIORITY] N-channel MOSFET (enhancement)
[PRIORITY] P-channel MOSFET (enhancement)	N-channel MOSFET (depletion)	P-channel MOSFET (depletion)
[PRIORITY] N-channel power MOSFET	P-channel power MOSFET	[PRIORITY] IGBT (insulated gate bipolar transistor)
GaN HEMT	SiC MOSFET	FinFET
FDSOI transistor	Dual-gate MOSFET	LDMOS
VDMOS		

Thyristors & Power Semiconductors (11 components)

[PRIORITY] SCR (silicon controlled rectifier)	[PRIORITY] Triac	GTO (gate turn-off thyristor)
MCT (MOS-controlled thyristor)	PUT (programmable unijunction transistor)	[PRIORITY] UJT (unijunction transistor)
SCS (silicon controlled switch)	LASCR (light-activated SCR)	SBS (silicon bilateral switch)
SIDAC	SUS (silicon unilateral switch)	

Op-Amps & Comparators (21 components)

[PRIORITY] Op-amp (generic)	[PRIORITY] Inverting amplifier	[PRIORITY] Non-inverting amplifier
[PRIORITY] Voltage follower / buffer	[PRIORITY] Comparator	[PRIORITY] Schmitt trigger
[PRIORITY] Instrumentation amplifier	Differential amplifier	Summing amplifier
Integrator (op-amp)	Differentiator (op-amp)	Transconductance amplifier (OTA)
Log amplifier	Antilog amplifier	Sample-and-hold amplifier
[PRIORITY] Transimpedance amplifier (TIA)	Charge amplifier	[EXISTS] Voltage-controlled voltage source (VCVS)
[EXISTS] Current-controlled voltage source (CCVS)	[EXISTS] Voltage-controlled current source (VCCS)	[EXISTS] Current-controlled current source (CCCS)

Logic Gates & Combinational ICs (22 components)

[PRIORITY] AND gate	[PRIORITY] OR gate	[PRIORITY] NOT gate (inverter)
[PRIORITY] NAND gate	[PRIORITY] NOR gate	[PRIORITY] XOR gate
[PRIORITY] XNOR gate	[PRIORITY] Buffer gate	[PRIORITY] Tri-state buffer
[PRIORITY] Half adder	[PRIORITY] Full adder	4-bit ripple carry adder
[PRIORITY] Multiplexer (MUX) 2:1	Multiplexer 4:1	Multiplexer 8:1

[PRIORITY] Demultiplexer (DEMUX)	[PRIORITY] Encoder (priority)	Decoder (3-to-8)
[PRIORITY] BCD-to-7-segment decoder	[PRIORITY] Comparator (4-bit)	Parity generator/checker
Look-ahead carry unit		

Sequential Logic & Flip-Flops (21 components)

[PRIORITY] SR latch	[PRIORITY] SR flip-flop (clocked)	[PRIORITY] D flip-flop
[PRIORITY] JK flip-flop	[PRIORITY] T flip-flop	[PRIORITY] D latch
Master-slave JK flip-flop	[PRIORITY] 4-bit binary counter	BCD counter
Ring counter	Johnson counter	[PRIORITY] Up/down counter
Shift register (SISO)	[PRIORITY] Shift register (SIPO)	Shift register (PISO)
Shift register (PIPO)	Universal shift register	LFSR (linear feedback shift register)
Sequence detector (Mealy)	Sequence detector (Moore)	RAM cell (static)

Timers & Oscillators (17 components)

[PRIORITY] 555 timer IC	556 dual timer	558 quad timer
[PRIORITY] Astable multivibrator	[PRIORITY] Monostable multivibrator	Bistable multivibrator
Colpitts oscillator	Hartley oscillator	Clapp oscillator
[PRIORITY] Pierce oscillator (crystal)	Wien bridge oscillator	RC phase-shift oscillator
Ring oscillator	Voltage-controlled oscillator (VCO)	[PRIORITY] PLL (phase-locked loop)
Clock generator IC	Relaxation oscillator	

Analog ICs & Regulators (25 components)

[PRIORITY] Voltage regulator (78xx linear)	Voltage regulator (79xx negative)	[PRIORITY] LDO (low-dropout) regulator
[PRIORITY] Adjustable regulator (LM317)	[PRIORITY] Shunt regulator (TL431)	[PRIORITY] Buck converter (switching)
[PRIORITY] Boost converter	Buck-boost converter	Flyback converter
Forward converter	Cuk converter	SEPIC converter
Charge pump	Voltage multiplier (Cockcroft-Walton)	[PRIORITY] DAC (digital-to-analog converter)
[PRIORITY] ADC (analog-to-digital converter)	Sample and hold IC	Multiplier IC (analog)
RMS-to-DC converter	Isolation amplifier	[PRIORITY] Audio power amplifier IC
Class-D amplifier IC	Variable gain amplifier (VGA)	Active filter IC (UAF42)

[PRIORITY] Analog switch IC
(4066)

Microcontrollers & Processors (16 components)

[PRIORITY] Microcontroller (generic block)	[PRIORITY] Arduino Uno (ATmega328P)	ATtiny microcontroller
[PRIORITY] PIC microcontroller	[PRIORITY] STM32 (ARM Cortex-M)	[PRIORITY] ESP32
ESP8266	Raspberry Pi Pico (RP2040)	[PRIORITY] 8051 microcontroller
AVR generic	ARM Cortex-A (MPU block)	DSP processor block
[PRIORITY] FPGA block	CPLD block	RISC-V processor block
System-on-chip (SoC) generic block		

Memory ICs (16 components)

[PRIORITY] SRAM IC	DRAM IC	SDRAM IC
[PRIORITY] Flash memory IC	[PRIORITY] EEPROM IC	EPROM IC
ROM IC	PROM IC	NVRAM IC
MRAM IC	FRAM (ferroelectric RAM)	SPI Flash (e.g. W25Q32)
[PRIORITY] I2C EEPROM (24Cxx)	[PRIORITY] SD card slot (symbol)	NOR Flash
NAND Flash		

Interface & Communication ICs (23 components)

[PRIORITY] UART transceiver	[PRIORITY] SPI interface block	[PRIORITY] I2C interface block
[PRIORITY] RS-232 line driver (MAX232)	[PRIORITY] RS-485 transceiver	CAN bus transceiver
LIN transceiver	USB 2.0 controller	USB Type-C controller
[PRIORITY] Ethernet PHY IC	Ethernet MAC	I2S audio interface
[PRIORITY] Level shifter (logic)	Bus buffer (74-series)	[PRIORITY] Opto-isolator / optocoupler
Digital isolator IC	[PRIORITY] Bluetooth module (HC-05)	Wi-Fi module (ESP8266)
Zigbee module	LoRa module	NFC controller IC
GPS module block	CAN controller (MCP2515)	

Sensors & Transducers (37 components)

[PRIORITY] Temperature sensor (LM35)	[PRIORITY] Thermistor (NTC)	[PRIORITY] RTD (Pt100/Pt1000)
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[PRIORITY] Thermocouple (K/J type)	[PRIORITY] Pressure sensor	Force sensor (load cell)
Strain gauge	[PRIORITY] Piezoelectric sensor	[PRIORITY] Accelerometer (ADXL345)
[PRIORITY] Gyroscope (MPU-6050)	[PRIORITY] IMU (6/9 DOF)	Magnetometer / compass
[PRIORITY] Hall effect sensor	[PRIORITY] Current sensor (ACS712)	[PRIORITY] Voltage sensor divider
[PRIORITY] Humidity sensor (DHT22)	[PRIORITY] Gas sensor (MQ-x series)	UV sensor
[PRIORITY] Infrared sensor (IR receiver)	[PRIORITY] Ultrasonic sensor (HC-SR04)	LIDAR sensor block
[PRIORITY] Capacitive touch sensor	Proximity sensor	Color sensor (TCS3200)
[PRIORITY] Sound / microphone (MEMS)	LVDT (linear variable differential transformer)	[PRIORITY] Encoder (rotary optical)
Resolver	Flow sensor	Level sensor
[PRIORITY] Reed switch	Tilt sensor	Rain sensor
Flex sensor	Soil moisture sensor	pH sensor
Turbidity sensor		

Displays & Indicators (16 components)

[PRIORITY] LED indicator	[PRIORITY] 7-segment display (single)	7-segment display (4-digit)
[PRIORITY] 16x2 LCD (HD44780)	20x4 LCD	[PRIORITY] OLED display block (SSD1306)
E-ink display block	TFT LCD block	VFD (vacuum fluorescent display)
Nixie tube	Bar graph LED array	[PRIORITY] RGB LED
WS2812B addressable LED	[PRIORITY] Dot matrix LED (8x8)	Laser pointer / laser module
Plasma display block		

Actuators & Electromechanical (20 components)

[PRIORITY] DC motor	[PRIORITY] AC motor (single-phase)	[PRIORITY] AC motor (3-phase)
[PRIORITY] Stepper motor	[PRIORITY] Servo motor	BLDC motor
[PRIORITY] Solenoid	[PRIORITY] Relay (SPDT)	Relay (DPDT)
[PRIORITY] Solid-state relay (SSR)	Contactors	[PRIORITY] Buzzer (active)
Buzzer (passive)	[PRIORITY] Speaker / loudspeaker	[PRIORITY] Piezo buzzer
Electromagnetic lock	Linear actuator	Vibration motor
[PRIORITY] H-bridge (L298N)	H-bridge (L293D)	

Switches, Relays & Push-buttons (21 components)

[EXISTS] SPST switch	[PRIORITY] SPDT switch	DPST switch
[PRIORITY] DPDT switch	[PRIORITY] Push-button (NO)	[PRIORITY] Push-button (NC)
Momentary push button	[PRIORITY] Toggle switch	Rotary switch
[PRIORITY] DIP switch	[PRIORITY] Slide switch	Joystick (2-axis) symbol
Tact switch	Foot switch	Magnetic reed switch
[PRIORITY] Microswitch (limit switch)	Pressure switch	Float switch
Thermal switch (bimetal)	Key switch	Capacitive touch switch

Power Sources & Batteries (18 components)

[EXISTS] DC voltage source	[PRIORITY] AC voltage source	[EXISTS] Current source
[PRIORITY] Battery (single cell)	[PRIORITY] Battery (multi-cell)	Li-ion cell
LiPo cell	NiMH cell	Lead-acid battery (symbol)
Fuel cell	[PRIORITY] Solar cell	Supercapacitor (power)
[PRIORITY] Mains supply terminal (230V AC)	3-phase supply terminal	[PRIORITY] USB power connector symbol
[PRIORITY] Power jack symbol	Wireless charging coil	Piezoelectric energy harvester

Connectors & Terminals (26 components)

[PRIORITY] Wire junction dot	[PRIORITY] Terminal / test point	[PRIORITY] Pin header (male)
Pin header (female)	[PRIORITY] 2-pin connector	4-pin connector
JST connector	DB9 (serial) connector	DB25 connector
[PRIORITY] RJ45 (Ethernet) jack	USB-A connector	USB micro-B connector
USB Type-C connector	BNC connector	SMA connector
Banana jack / binding post	[PRIORITY] Screw terminal	DIP IC socket
SOIC footprint	[PRIORITY] Power plane (+VCC)	[EXISTS] Ground symbol
Earth ground	Chassis ground	Digital ground
Coax cable symbol	Twisted pair symbol	

Protection & Conditioning (18 components)

[PRIORITY] Fuse (slow-blow)	Fuse (fast-blow)	Circuit breaker (symbol)
[PRIORITY] TVS diode array (ESD protection)	Surge protector block	Inrush current limiter (NTC thermistor)
[PRIORITY] EMI filter (LC)	Common-mode filter	pi (pi) filter
T-filter	[PRIORITY] Isolation transformer	Crowbar circuit

Overvoltage protection IC	Reverse polarity protection (MOSFET)	Resettable fuse (PPTC)
Gas discharge tube (GDT)	[PRIORITY] Metal-oxide varistor (MOV)	Zener clamp circuit

RF, Antenna & Wireless (28 components)

[PRIORITY] Antenna (generic)	Dipole antenna	Monopole antenna
Patch antenna	Helical antenna	Yagi-Uda antenna
Loop antenna	Parabolic reflector	RF amplifier block
Low-noise amplifier (LNA)	Power amplifier (PA)	Bandpass filter (RF)
Low-pass filter (RF)	High-pass filter (RF)	Duplexer
Circulator	Isolator	Directional coupler
RF mixer	RF modulator / demodulator	FM/AM receiver IC block
Superheterodyne block	RF switch	SAW filter
BAW filter	VCO (voltage-controlled oscillator)	Frequency synthesizer IC
RF balun / transformer		

Filters & Signal Processing (22 components)

[PRIORITY] RC low-pass filter	[PRIORITY] RC high-pass filter	RL low-pass filter
RL high-pass filter	[PRIORITY] LC bandpass filter	LC notch filter
[PRIORITY] Active low-pass (Sallen-Key)	Active high-pass (Sallen-Key)	Active bandpass
Active notch (twin-T)	State-variable filter	Biquad filter
Butterworth filter block	Chebyshev filter block	Bessel filter block
Elliptic filter block	FIR filter block	IIR filter block
Integrator	Differentiator	Delay line
Sample rate converter		

Power Electronics & Drives (21 components)

[PRIORITY] Half-wave rectifier	[PRIORITY] Full-wave rectifier	[PRIORITY] Bridge rectifier (Graetz)
3-phase bridge rectifier	Voltage doubler	Voltage tripler
Chopper (DC-DC)	[PRIORITY] Inverter (DC-AC)	Single-phase inverter
3-phase inverter bridge	[PRIORITY] PWM generator	[PRIORITY] Gate driver IC
Bootstrap gate driver	Snubber circuit (RC)	Resonant converter
[PRIORITY] Variable frequency drive (VFD) block	Soft starter block	AC voltage controller (TRIAC-based)
Cycloconverter block	Matrix converter block	Switched reluctance drive block

Measurement & Test Instruments (15 components)

[PRIORITY] Ammeter	[PRIORITY] Voltmeter	Ohmmeter
Wattmeter	Oscilloscope probe	Multimeter symbol
Signal generator symbol	Frequency counter symbol	Power meter
LCR meter	Logic analyser symbol	Network analyser symbol
Spectrum analyser symbol	Current probe	Differential probe

Electroacoustic & Audio (15 components)

[PRIORITY] Microphone (dynamic)	Microphone (condenser)	MEMS microphone
[PRIORITY] Speaker (loudspeaker)	Headphone symbol	Earpiece
[PRIORITY] Piezo transducer	Ultrasonic transducer	Audio transformer
[PRIORITY] Audio jack (3.5 mm)	Volume control (pot)	Tone control block
Equaliser block	Crossover network	Audio codec IC (PCM5102)

Optoelectronics & Photonics (16 components)

[PRIORITY] Photodiode	[PRIORITY] Phototransistor	[PRIORITY] Photoresistor / LDR
[PRIORITY] Optocoupler / opto-isolator	[PRIORITY] LED (generic)	[PRIORITY] IR LED
[PRIORITY] Laser diode	VCSEL	[PRIORITY] Solar panel (multi-cell)
Photodetector (APD)	CCD image sensor block	CMOS image sensor block
Fiber-optic transmitter	Fiber-optic receiver	Optical isolator (fiber)
Light pipe symbol		

MEMS, Emerging & Nanotechnology (15 components)

MEMS accelerometer	MEMS gyroscope	MEMS microphone
MEMS pressure sensor	MEMS resonator	Micro-heater
Micro-fuel cell	Piezoelectric energy harvester	Thermoelectric generator (TEG)
Thermopile sensor	CNT (carbon nanotube) FET	Graphene FET
Quantum dot LED	Single-electron transistor	Quantum bit (qubit) symbol

High Voltage & Electrical Power (23 components)

[PRIORITY] Transformer (power, 1-phase)	Transformer (3-phase)	Auto-transformer
[PRIORITY] CT (current transformer)	PT (potential transformer)	Circuit breaker (HV)
Isolator switch (disconnect)	Lightning arrester	Surge arrester

Busbar	Power cable (symbol)	Switchgear block
Capacitor bank	Reactor (shunt / series)	STATCOM block
HVDC converter block	Arc furnace block	Distribution transformer
[PRIORITY] Generator (AC)	Generator (DC)	Synchronous motor
[PRIORITY] Induction motor	Motor starter block	

Development Boards & Modules —

PART 2

Skeletal Pin Reference

PWR	GND	GPIO	PWM~	ADC	UART	SPI	I2C	Special
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Arduino Family

Arduino Uno R3

ATmega328P | 5V | 32KB Flash | 14D / 6A

The most-taught board worldwide. Standard for school labs.

VIN	5V	3.3V	GND x3
IOREF	RESET	AREF	D0 / RX
D1 / TX	D2 INT0	D3~ INT1	D4
D5~	D6~	D7	D8
D9~	D10~ SS	D11~ MOSI	D12 MISO
D13 SCK / LED	A0	A1	A2
A3	A4 / SDA	A5 / SCL	

Arduino Nano

ATmega328P | 5V | 32KB Flash | 14D / 8A

Compact breadboard-friendly version of Uno. Very popular in mini-projects.

VIN	5V	3.3V	GND x4
RESET	AREF	D0/RX	D1/TX
D2 INT0	D3~	D4	D5~
D6~	D7	D8	D9~
D10~ SS	D11~ MOSI	D12 MISO	D13 SCK/LED
A0	A1	A2	A3
A4/SDA	A5/SCL	A6 (analog only)	A7 (analog only)

Arduino Mega 2560

ATmega2560 | 5V | 256KB Flash | 54D / 16A

Big-pin board for complex projects. Robotics favourite.

VIN	5V	3.3V	GND x multiple
IOREF	RESET	D0/RX0	D1/TX0
D14/TX3 D15/RX3	D16/TX2 D17/RX2	D18/TX1 + INT5	D19/RX1 + INT4
D20/SDA + INT3	D21/SCL + INT2	D2-D13~ (PWM)	D22-D53 (Digital I/O)
D50 MISO	D51 MOSI	D52 SCK	D53 SS
A0-A15 (16 analog inputs)	AREF		

Arduino Nano Every

ATmega4809 | 5V | 48KB Flash | 14D / 8A

Modern Nano replacement with more memory and better peripherals.

VIN	5V	3.3V	GND
D0/RX	D1/TX	D2 INT	D3~
D4	D5~	D6~	D7
D8	D9~	D10~ SS	D11~ MOSI
D12 MISO	D13 SCK/LED	A0-A7 (10-bit)	A4/SDA
A5/SCL			

Arduino Leonardo

ATmega32U4 | 5V | 32KB Flash | 20D / 12A

Native USB HID — can act as keyboard or mouse. USB projects.

VIN	5V	3.3V	GND
D0/RX	D1/TX	D2 INT1/SDA	D3~ INT0/SCL
D4 A6	D5~	D6~ A7	D7 INT4
D8 A8	D9~ A9	D10~ A10	D11~
D12 A11	D13~ SCK/LED	A0-A5	SDA/SCL
MOSI/MISO/SS			

Arduino Pro Mini

ATmega328P | 3.3V or 5V | 32KB | 14D / 8A

No USB, tiny form factor. Space-constrained embedded builds.

VCC (3.3V or 5V)	GND	RAW (unregulated)	RESET
D0/RX	D1/TX	D2 INT0	D3~ INT1
D4	D5~	D6~	D7
D8	D9~	D10~ SS	D11~ MOSI
D12 MISO	D13 SCK	A0-A3	A4/SDA
A5/SCL	A6 (analog only)	A7 (analog only)	

ESP Family (WiFi / Bluetooth)

ESP32 DevKit V1

ESP32 Xtensa LX6 Dual-core | 3.3V | 4MB Flash | 30 GPIO | WiFi + BT

Most popular IoT board. Dual-core, WiFi + Bluetooth. Used heavily in college IoT labs.

VIN / 5V	3.3V	GND x4	EN
BOOT / GPIO0	GPIO1 / TXD0	GPIO3 / RXD0	GPIO16 / RX2
GPIO17 / TX2	GPIO21 / SDA	GPIO22 / SCL	GPIO18 / SCK
GPIO19 / MISO	GPIO23 / MOSI	GPIO5 / SS	GPIO2 (LED/boot)
GPIO4	GPIO12-15	GPIO25 DAC1	GPIO26 DAC2
GPIO32-39 ADC1 (12-bit)	GPIO34-39 (input only)	Touch pins GPIO 0,2,4,12-15,27,32,33	

ESP32-CAM

ESP32-S + OV2640 2MP Camera | 3.3V/5V | 4MB Flash | 10 usable GPIO

ESP32 with 2MP camera. Face detection, surveillance, video streaming projects.

5V	GND	3.3V	GPIO0 (boot/flash)
GPIO1/TXD	GPIO3/RXD	GPIO12-15 (SD card SPI)	GPIO16 PSRAM CS
GPIO2 LED flash	GPIO4 Onboard LED	CAM: Y2-Y9, VSYNC, HREF	CAM: PCLK, XCLK, SIOD, SIOC

NodeMCU ESP8266

ESP8266 | 3.3V | 4MB Flash | 11 GPIO | WiFi only

Classic WiFi-only board. Great intro to IoT. Very cheap and widely available.

VIN / 3.3V	GND	D0 / GPIO16 (no INT)	D1 / GPIO5 SCL
D2 / GPIO4 SDA	D3 / GPIO0 (boot)	D4 / GPIO2 (LED/boot)	D5 / GPIO14 SCK
D6 / GPIO12 MISO	D7 / GPIO13 MOSI	D8 / GPIO15 SS	RX / GPIO3
TX / GPIO1	A0 (0-1V single analog)		

Wemos D1 Mini (ESP8266)

ESP8266 | 3.3V | 4MB Flash | 11 GPIO | WiFi | Shield ecosystem

Tiny ESP8266 board with shield ecosystem. Compact IoT nodes.

5V	3.3V	GND	D0/GPIO16
D1/GPIO5 SCL	D2/GPIO4 SDA	D3/GPIO0 (boot)	D4/GPIO2 (LED)
D5/GPIO14 SCK	D6/GPIO12 MISO	D7/GPIO13 MOSI	D8/GPIO15 SS
RX/GPIO3	TX/GPIO1	A0 (0-3.2V via divider)	

Raspberry Pi Family

Raspberry Pi 4 / 5

BCM2711 Cortex-A72 | 3.3V GPIO | SD card | 40-pin header | WiFi + BT

Full Linux SBC. College final-year projects, AI/ML, robotics, vision systems.

Pin 1: 3.3V	Pin 2/4: 5V	GND (pins 6,9,14,20,25,30,34,39)	GPIO2 / SDA1 (Pin 3)
GPIO3 / SCL1 (Pin 5)	GPIO4 (Pin 7)	GPIO17 (Pin 11)	GPIO27 (Pin 13)
GPIO22 (Pin 15)	GPIO10 / MOSI (Pin 19)	GPIO9 / MISO (Pin 21)	GPIO11 / SCLK (Pin 23)
GPIO8 / CE0 (Pin 24)	GPIO7 / CE1 (Pin 26)	GPIO14 / TXD (Pin 8)	GPIO15 / RXD (Pin 10)
GPIO18~ PWM0 (Pin 12)	GPIO19~ PWM1 (Pin 35)	GPIO5,6,12,13,16,23,24,25,26	ID_SD / ID_SC (HAT EEPROM)

Raspberry Pi Pico / Pico W

RP2040 Dual Cortex-M0+ | 3.3V | 2MB Flash | 26 GPIO | (Pico W: WiFi)

Microcontroller Pi. Cheap, MicroPython ready. Rising star in school labs.

VSYS Pin39 (1.8-5.5V)	VBUS Pin40 (USB 5V)	3V3 Pin36	GND x8
3V3_EN Pin37	RUN Pin30 (reset)	GP0 / TX UART0	GP1 / RX UART0
GP4 / SDA I2C0	GP5 / SCL I2C0	GP2 / SCK SPI0	GP3 / TX SPI0 MOSI
GP17 SPI0 CSn	GP0-GP22 (all PWM capable)	GP26 ADC0	GP27 ADC1
GP28 ADC2	GP29 ADC3 (VSYS/3)	BOOTSEL	

Raspberry Pi Zero 2W

BCM2710A1 Cortex-A53 | 3.3V GPIO | SD card | 40-pin | WiFi + BT

Tiny Linux Pi. Identical 40-pin GPIO as Pi 4. Wearable/IoT projects.

Identical 40-pin header to RPi 4	3.3V (Pin 1), 5V (Pin 2/4)	GND (multiple)	GPIO2/SDA, GPIO3/SCL
GPIO14/TXD, GPIO15/RXD	GPIO10/MOSI, GPIO9/MISO, GPIO11/SCLK, GPIO8/CE0	GPIO18~ PWM0, GPIO19~ PWM1	GPIO4-27 general

Raspberry Pi Pico 2

RP2350 Dual Cortex-M33 | 3.3V | 4MB Flash | 26 GPIO

Updated Pico with faster core and more memory. New in college labs.

Same pinout as Pico	GP0-GP22 (all PWM)	GP26-GP28 ADC	GP0/GP1 UART0
GP4/GP5 I2C0	GP2-GP5 SPI0	BOOTSEL	

STM32 / ARM Family

STM32 Blue Pill

STM32F103C8T6 Cortex-M3 | 3.3V | 64KB Flash | 37 GPIO

Cheap ARM Cortex-M3. Popular in college embedded labs replacing Arduino.

VCC 3.3V	VIN 5V (USB)	GND	RESET
BOOT0 (HIGH=flash)	PA0-PA7 ADC + TIM2 PWM	PA8-PA10 TIM1 PWM	PA9/TX1
PA10/RX1	PA11 USB D-	PA12 USB D+	PA13/SWDIO
PA14/SWCLK	PB6/SCL1	PB7/SDA1	PB3/SCK
PB4/MISO	PB5/MOSI	PA4/NSS	PB0-PB15 GPIO
PC13 (LED)			

STM32 Nucleo-64

STM32F446RE Cortex-M4 | 3.3V | 512KB Flash | 76 GPIO

Official ST dev board. Arduino-compatible + ST morpho header. DSP labs.

VIN 7-12V	5V	3.3V	GND
Arduino header D0-D15	Arduino header A0-A5	SDA/SCL (Arduino)	D10-D13 SPI (Arduino)
D0/D1 UART (Arduino)	ST Morpho CN7/CN10 (full 76 GPIO)	USER LED PA5	USER BTN PC13
SWD header (ST-Link V2.1)			

BBC micro:bit & Beginner Boards

BBC micro:bit V2

nRF52833 Cortex-M4 | 3.3V | 512KB Flash | 5 large + 19 edge pins | BLE

UK school standard. Built-in sensors, LED matrix, BLE. Perfect for beginners 10+.

P0 (large pad, touch, analog)	P1 (large pad, touch, analog)	P2 (large pad, touch, analog)	3V (large pad)
GND (large pad)	P3 LED col1	P4 LED col2 / ADC	P5 Button A
P11 Button B	P6/P7/P9/P10 LED lines	P8/P12 GPIO	P13/SCK
P14/MISO	P15/MOSI	P16 CS	P19/SCL (internal sensors)
P20/SDA	Onboard: 5x5 LED matrix	Onboard: accelerometer + compass	Onboard: temperature sensor
Onboard: speaker + mic (V2)	Onboard: BLE 5.0 radio		

NVIDIA Jetson Nano

Cortex-A57 + 128-core Maxwell GPU | 5V | 16GB eMMC | 40-pin GPIO

AI/vision board for college final year. GPIO identical to Pi, adds GPU.

Pin 1: 3.3V	Pin 2/4: 5V	GND x multiple	40-pin GPIO same as RPi 4
SDA/SCL I2C0 + I2C1	TXD/RXD UART	SPI0: MOSI/MISO/SCK/CE0/CE1	PWM0 GPIO12, PWM1 GPIO13
CSI camera x2, DSI display	M.2 Key-E (WiFi optional)		

Adafruit / SparkFun / Maker

Adafruit Feather M0

ATSAMD21G18 Cortex-M0+ | 3.3V | 256KB Flash | 20 GPIO | Wing ecosystem

Thin board with LiPo connector built-in. Wing shield ecosystem.

3V	VBAT (LiPo)	VUSB	GND
RESET	A0 (also DAC out)	A1-A5	D0/RX
D1/TX	D5~,D6~,D9~,D10~,D11~,D12~,D13~	SDA/SCL	MOSI/MISO/SCK/SS

Teensy 4.1

IMXRT1062 Cortex-M7 600MHz | 3.3V | 8MB Flash | 55 GPIO

Fastest Arduino-compatible. Audio, USB host, high-speed comms.

VIN 3.6-5.5V	3.3V	GND	D0/RX1
D1/TX1	TX/RX x8 ports	D2-D15~ + more (32 PWM)	A0-A17 (12-bit ADC)
SDA0/SCL0, SDA1/SCL1, SDA2/SCL2 (3 I2C)	MOSI/MISO/SCK/CS x3 (3 SPI)	Built-in SD card slot	USB host pins

SparkFun RedBoard Artemis

Apollo3 Blue Cortex-M4F | 3.3V | 1MB Flash | 21 GPIO | BLE

Ultra low-power BLE + ML support. Popular in US university labs.

VIN 3.3-6V	3.3V	GND	D0/RX
D1/TX	D2-D13~	A0-A5	SDA/SCL
MOSI/MISO/SCK/CS	PDM onboard MEMS mic (ML)		

Wireless & Popular Modules

HC-05 Bluetooth Module

BC417 | 3.3V-5V | Classic BT 2.0

Serial Bluetooth. Pairs with phones/PC. Wireless control projects.

VCC (3.6-6V)	GND	TXD (to MCU RX)	RXD (use voltage divider!)
KEY/EN (HIGH = AT mode)	STATE (blinks=not paired)		

HC-SR04 Ultrasonic Sensor

– | 5V | 4 pins

4-pin distance sensor. Teaches timing & interrupts. Most-used school component.

VCC (5V)	GND	TRIG (send 10us HIGH)	ECHO (pulse width = distance)
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NRF24L01+ Module

nRF24L01+ | 3.3V only | 2.4 GHz radio

2.4 GHz radio for wireless sensor networks. Multi-node comms.

VCC 3.3V only!	GND	CE (chip enable)	CSN (chip select NOT)
SCK	MOSI	MISO	IRQ (interrupt, optional)

SIM800L GSM Module

SIM800L | 3.7-4.2V (LiPo!) | GSM/GPRS

Send SMS, calls, GPRS. Final-year IoT favourite.

VCC 3.7-4.2V (LiPo NOT 5V!)	GND	TXD (to MCU RX)	RXD (from MCU TX)
RST (pull LOW to reset)	NET LED (blink=searching)		

FPGA / CPLD (Advanced Students)

Basys 3 (Xilinx Artix-7)

XC7A35T | 3.3V | 106 GPIO | University digital design standard

Switches, LEDs, 7-seg display, VGA, USB-UART. Standard Verilog/VHDL board.

USB JTAG programming	VCC / GND rails	Pmod JA/JB/JC/JXADC (4x12 Pmod)	16 slide switches SW0-SW15
16 LEDs LD0-LD15	5 push buttons BTNC/UD/L/R	4-digit 7-segment display	VGA port
USB-UART (FTDI)	100 MHz oscillator		

DE0-Nano (Intel Cyclone IV)

EP4CE22F17C6 | 3.3V | 72 GPIO | Common in Indian EE colleges

50 GPIO, 8 ADC channels, Arduino-style header. Verilog/VHDL labs.

JTAG (USB Blaster)	VCC3.3 / VCC5 / GND	GPIO-0 (26 pins)	GPIO-1 (26 pins)
8 LEDs	4 push buttons	4 DIP switches	8 ADC channels (ADC128S022)
2x13 Arduino-style header	50 MHz clock		